

# Riyadh Mobility Intelligence Starter Kit

## Build and Deployment Workbook

A polished, local-first build path for teams adapting mobility intelligence, district scoring, and Azure-backed urban development prototypes.

### Document targets

- **Run locally**  
Sample data, map fallback, district selector, score panel
- **Explain the system**  
FastAPI, JavaScript, data fallback, Azure services
- **Deploy credibly**  
Container Apps, Maps, Blob, Cosmos, App Insights
- **Adapt by track**  
Technology, People, Sustainability, Culture

RIYADH URBAN INTELLIGENCE LAB — 2026

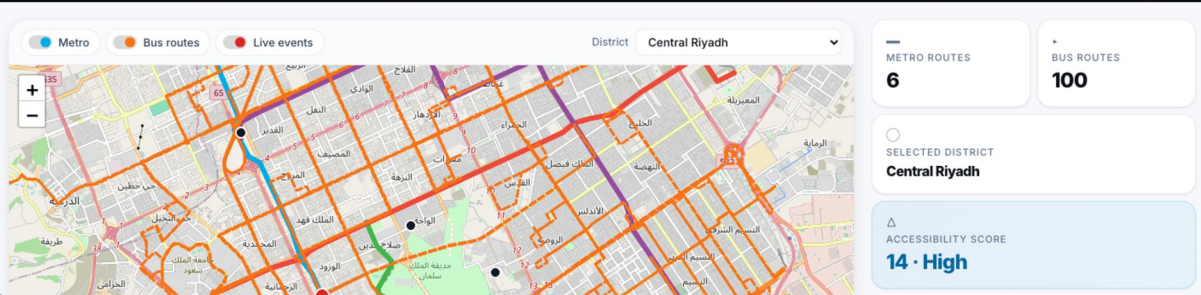
## Mobility Intelligence

لوحة التنقل الذكي

6 real metro lines & 100+ bus routes from RCRC open data. Select any Riyadh district to compute a transit accessibility score — stored in Azure Cosmos DB, served by Azure Container Apps.

**6** METRO LINES  
**100** BUS ROUTES  
**10** DISTRICTS  
**RCRC** OPEN DATA

METRO LINES (REAL RCRC DATA): Blue — SAB Bank ↔ Ad Dar Al Baida Red — KSU ↔ King Fahad Sport City Orange — Jeddah Rd ↔ Khashm Al An Yellow — Airport ↔ KAFD Green — Ministry of Ed ↔ Nat'l Museum Purple —



Transformational Technology

Prosperous People

Azure-backed path

Sample-first fallback

Prepared for atomcamp Arabia and Microsoft collaboration

Repository, local app, deployed URL, and Azure resources are listed inside the workbook.

# How to Use This Guide

A practical build path for teams who need to understand, run, inspect, deploy, and adapt the Riyadh Mobility Intelligence starter kit with minimal help.

## Quick Access

|                                                                                   |                       |                                                                                                                                       |
|-----------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>GitHub repo</b>    | <a href="https://github.com/Esturban/microsoft-hackathon-riyadh-mobility">github.com/Esturban/microsoft-hackathon-riyadh-mobility</a> |
|  | <b>Local app</b>      | <a href="http://127.0.0.1:8000">http://127.0.0.1:8000</a>                                                                             |
|  | <b>Deployed app</b>   | Paste WEB_APP_URL from azd output here                                                                                                |
|  | <b>Resource index</b> | Appendix: Azure docs, deployment notes, project references                                                                            |

## Starter Kit, Not Final Model

Use this as a scaffold for credible prototypes. The included score is intentionally simple so builders can replace the data, rules, and Azure services as their track requires.

## Local-First Principle

Get the dashboard working from bundled sample data before adding cloud credentials, Blob Storage, Cosmos DB, or a deployed app URL.

## Builder Playbook

- 01 Understand app**  
Read what the dashboard, layers, score panel, and fallback chain are doing.
- 02 Run locally**  
Create the Python environment, start FastAPI, and wait for the slow initial load.
- 03 Inspect API**  
Check /health, /api/data-status, routes, districts, and score responses.
- 04 Select district**  
Use Riyadh North or Al Olaya and confirm the score panel updates.
- 05 Deploy Azure**  
Use az login, azd auth login, and scripts/deploy\_azure.sh when ready.
- 06 Adapt track**  
Keep the scaffold; replace data, scoring, layers, and Azure services by track.

## App layers to keep in view



Repository, local app, deployed URL, and Azure references are listed in the workbook appendix.

# Executive Build Brief

A local-first, Azure-ready workbook for teams turning open mobility data into a credible urban intelligence prototype.

**106**

route records across metro and bus sample layers

**10**

district records available for scoring and demos

**9**

API endpoints to inspect, extend, and deploy

**7+**

Azure services mapped to clear runtime jobs



## What this helps teams prove

A working dashboard can connect map layers, district context, explainable scoring, and cloud deployment without requiring every team to start from zero.



## What it deliberately does not claim

The score is a teachable proxy, not a formal transport model. Its value is the transparent scaffold: data in, API contract, map state, score explanation, cloud path.

## Build Path

**01**

### Run sample mode

Confirm local app, map fallback, route layers, district selector, and score panel.

**02**

### Inspect contracts

Review /health, /api/data-status, /api/routes, /api/districts, and /api/score.

**03**

### Connect Azure

Deploy Container Apps, Maps, Blob, Cosmos, monitoring, and diagnostics.

**04**

### Adapt for track

Replace data and scoring while keeping the proven frontend, API, and deployment shape.

## Design principle

Keep the first successful path simple: sample data locally, transparent API behavior, then Azure services only when the team is ready to make the prototype cloud-live.

Use this page as the one-minute orientation before the implementation sections.

# 1. What You Are Building

The application is a **Mobility Intelligence Starter Kit** for Riyadh. It lets a builder open a map, view public transport layers, select a district, and receive a transparent mobility access score. The same scaffold can later support district intelligence, sustainability overlays, culture routes, or other urban challenge ideas.

The starter kit includes:

- Riyadh metro lines and bus routes on a browser map.
- District selection and district-level mobility scoring.
- Mock live mobility events such as delay or congestion markers.
- Fallback-aware data loading that works without Azure credentials.
- Optional Azure-backed deployment using Azure Maps, Blob Storage, Cosmos DB, Container Apps, and Application Insights.
- A reusable pattern for teams that want to replace the data and scoring logic with another urban intelligence use case.

The product story is intentionally simple:

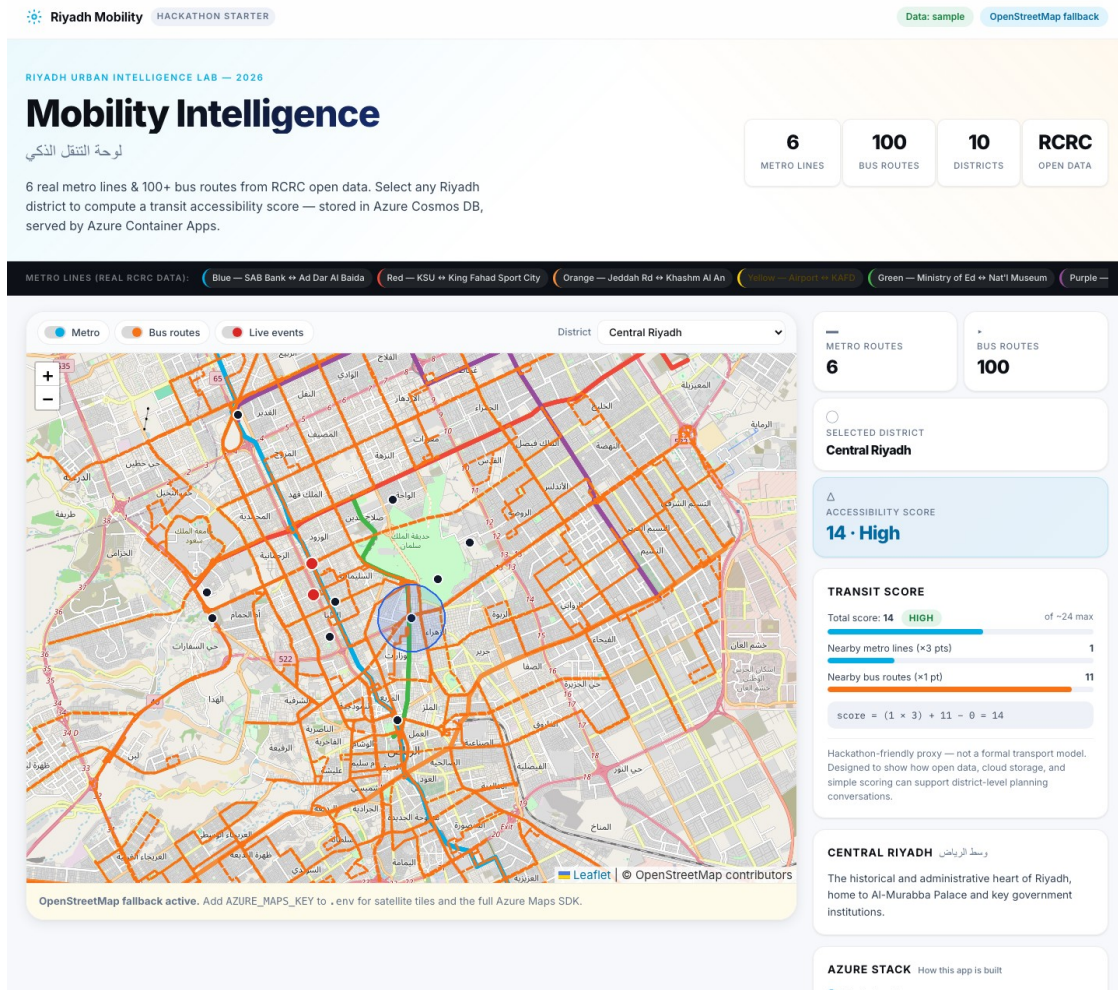
If a user selects a Riyadh district, can the app show nearby mobility infrastructure, explain route coverage, and turn that into a clear starter score?

The current score is a teachable proxy:

`score = (nearby metro count x 3) + nearby bus count - live delay penalty`

This formula is not the final planning answer. It is a clear starting point for showing how data, scoring, API design, map layers, and Azure services connect.





Local dashboard screenshot

Figure: local dashboard running from bundled sample data with metro lines, bus routes, district selector, score panel, and Azure service explainer visible.

## 2. Urban Challenge Track Mapping

The strongest primary fit is **Transformational Technology** because the current app already focuses on mobility visibility, route overlays, and cloud-backed intelligence. The strongest secondary fit is **Prosperous People** because the district scoring pattern can become a 15-minute city, walkability, or service-access score.

Sustainable Solutions and Culture should be treated as extension routes after the core app works.

Builder pitch:

Build a Riyadh mobility intelligence prototype with public route data, district scoring, Azure Maps, and an Azure-backed live path. Use it directly for Transformational Technology, or extend the same scaffold into Prosperous People, Sustainable Solutions, or Culture use cases.

# Services and Tools Matrix

The stack reads best as a set of clear jobs: local app runtime, data and maps, Azure hosting, and diagnostics. Each tool has one reason to exist.

## Local app runtime



### FastAPI

Backend API plus static frontend serving.

app/main.py, app/routes.py



### Vanilla JavaScript

Dashboard boot, map controls, panels, and API calls.

app/static/src/



### Sample files

Guaranteed fallback so teams can demo without Azure.

app/static/sample-data/

## Map and cloud data



### Azure Maps

Cloud map SDK and services when a key is configured.

AZURE\_MAPS\_KEY



### Blob Storage

Processed GeoJSON and JSON files for cloud data mode.

upload\_to\_blob.py



### Cosmos DB

Route, district, and event records for app-shaped data.

seed\_cosmos.py

## Deploy and observe



### Container Apps

Hosted FastAPI/frontend runtime.

azure.yaml, infra/main.bicep



### Container Registry

Container image storage for the deployed app.

infra/modules/



### App Insights + Logs

Health, failures, latency, requests, and diagnostics.

Azure portal

## Infrastructure workflow



### Bicep

Infrastructure-as-code for reproducible Azure resources.

infra/main.bicep



### Azure Developer CLI

Environment provisioning and deploy workflow.

azd / deploy script



### Tests

API, data-shape, health, and scoring checks.

tests/

Builder move: explain the services as responsibilities, not vendor names. The judge should hear runtime, map services, cloud data, monitoring, infrastructure, and local fallback.

# Architecture Blueprint

The system is designed to work locally first, then graduate to Azure-backed data, maps, and telemetry without changing the builder-facing workflow.



## Local-first path

Browser calls FastAPI. FastAPI serves the static frontend and API. The data layer reads bundled sample files so the app still works without Azure credentials.

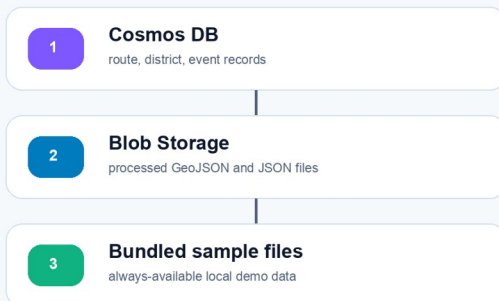
## Azure-backed path

The same app runs in Container Apps, uses Azure Maps when configured, reads Blob and Cosmos data when available, and reports telemetry to Application Insights.

## Fallback path

Cosmos DB is preferred when configured, Blob Storage is next, and bundled sample files are always the guaranteed floor for demos and local rebuilds.

## Fallback Chain



## Score Request Flow

- 1 User selects district
- 2 Frontend calls /api/score
- 3 Backend loads district/routes/events
- 4 Formula returns score + reasons
- 5 Panel renders explanation

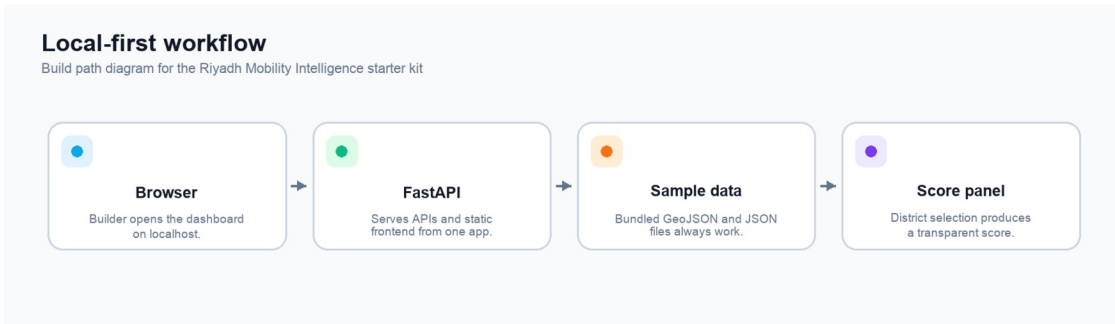
Key implementation rule: Azure should enhance the app, not be required for the first successful rebuild. That constraint keeps hackathon teams moving even when cloud setup is still in progress.

# 3. Architecture and Build Path

The app is designed to work locally first, then graduate into an Azure-backed live path. The core principle is simple: **the app must remain useful even when Azure credentials, remote services, or live data are unavailable.**

## Local-First Workflow

Local mode is the first build milestone. It should work on a builder laptop with no Azure account configured.

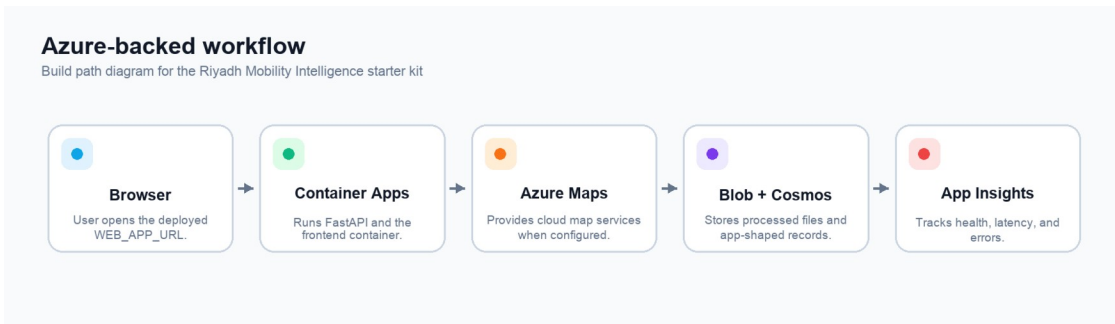


### Local-first workflow

Local mode uses the same UI and API shape that the cloud deployment uses. The difference is the data source: bundled sample files replace cloud-backed records.

## Azure-Backed Workflow

When deployed, the same containerized app runs in Azure Container Apps and can connect to Azure services for maps, storage, records, and monitoring.



### Azure-backed workflow

Use this path when the team is ready to show a credible cloud deployment or run post-deploy smoke tests.

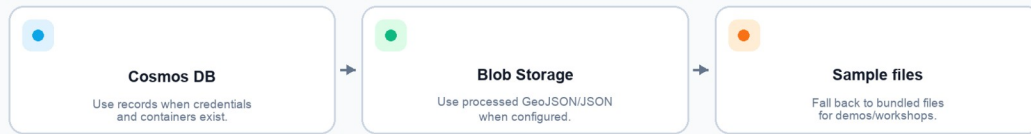
## Data Fallback Chain

The app should not fail just because cloud data is missing. Data access is intentionally fallback-aware.



## Data fallback chain

Build path diagram for the Riyadh Mobility Intelligence starter kit



### Data fallback chain

In practice, builders should start with `DATA_MODE=sample`, then move to Blob or Cosmos only after the local app is working.

## 4. Project Structure

Start with the app shell, API, frontend, sample data, deployment files, and tests.

riyadh-mobility-intelligence-dashboard/

|                  |                                            |
|------------------|--------------------------------------------|
| app/             |                                            |
| main.py          | # FastAPI app shell and static serving     |
| routes.py        | # API endpoints                            |
| data_access.py   | # sample, Blob, Cosmos data loading        |
| scoring.py       | # district mobility score                  |
| azure_clients.py | # Azure SDK client setup                   |
| config.py        | # environment variables and defaults       |
| static/          |                                            |
| index.html       | # single-page dashboard shell              |
| src/             | # frontend JavaScript and CSS              |
| sample-data/     | # always-on sample files                   |
| scripts/         | # fetch, normalize, upload, seed, validate |
| infra/           | # Bicep infrastructure modules             |
| docs/            | # build guide and supporting notes         |
| tests/           | # scoring, API, and data-shape checks      |
| azure.yaml       | # Azure Developer CLI project              |
| Dockerfile       | # container definition                     |
| requirements.txt | # Python dependencies                      |

| Area                | Start here                                      | Why it matters                                  |
|---------------------|-------------------------------------------------|-------------------------------------------------|
| Backend entrypoints | app/main.py, app/routes.py                      | FastAPI app shell, static serving, API contract |
| Data loading        | app/data_access.py, app/static/sample-data/     | sample, Blob, and Cosmos fallback behavior      |
| Score formula       | app/scoring.py                                  | transparent scoring logic that teams can adapt  |
| Frontend            | app/static/index.html, app/static/src/main.js   | page shell, boot sequence, API orchestration    |
| Map behavior        | app/static/src/map.js, app/static/src/layers.js | Azure Maps path, local fallback, overlays       |

| Area       | Start here                                            | Why it matters                              |
|------------|-------------------------------------------------------|---------------------------------------------|
| Deployment | azure.yaml, infra/main.bicep, scripts/deploy_azure.sh | cloud-live path                             |
| Tests      | tests/                                                | API, data-shape, health, and scoring checks |

## Where the Data Comes From

The bundled sample data is derived from Riyadh Commission for Riyadh City open geospatial datasets. The files live under `app/static/sample-data/` and keep the app useful even when no cloud services are configured.

| File                              | Contents                                                    | Records  |
|-----------------------------------|-------------------------------------------------------------|----------|
| riyadh_metro_lines_sample.geojson | Metro line features with names, colors, and geometry        | 6        |
| riyadh_bus_routes_sample.geojson  | Bus route features with route IDs and geometry              | 100      |
| district_centers_sample.geojson   | Riyadh district center points with English and Arabic names | 10       |
| mock_live_events_sample.json      | Sample delay or incident markers                            | variable |

Fresh data workflow:

```
python3 scripts/fetch_rcrc_data.py
python3 scripts/normalize_to_geojson.py
python3 scripts/validate_data.py
python3 scripts/upload_to_blob.py # optional cloud step
PYTHONPATH=. python3 scripts/seed_cosmos.py # optional cloud step
```

# Local Run Playbook

The fastest credible path is sample mode first: start the app, wait through slow load, verify the UI, then inspect API diagnostics.

01

## Clone

```
git clone <repo>
```

### Expected signal

folder exists

02

## Environment

```
python3 -m venv .venv
```

### Expected signal

virtual env created

03

## Install

```
pip install -r requirements.txt
```

### Expected signal

FastAPI + test deps installed

04

## Configure

```
cp .env.example .env
```

### Expected signal

DATA\_MODE=sample

05

## Start

```
python -m uvicorn app.main:app --reload
```

### Expected signal

server at 127.0.0.1:8000

06

## Verify

```
/health and /api/data-status
```

### Expected signal

ok + sample mode

### Slow-load rule

Wait for district selector, layer controls, map area, and score panel. External tiles may lag; overlays and panels are the success signal.

### Local success state

Dashboard loads, metro and bus layers appear, selector lists 10 districts, score panel updates, and data status reports sample mode.

Builder move: do not add Azure complexity until this page passes. Local reliability is the foundation for every credible demo, deployment, and track adaptation.

## 5. Run Locally

Local setup is deliberately direct. The goal is to get one useful screen running before adding cloud services.

### Play-by-Play

```
git clone https://github.com/Esturban/microsoft-hackathon-riyadh-mobility.git
cd microsoft-hackathon-riyadh-mobility
python3 -m venv .venv
source .venv/bin/activate # Windows: .venv\Scripts\activate
pip install -r requirements.txt
cp .env.example .env
```

Open `.env` and confirm the default local mode:

`DATA_MODE=sample`

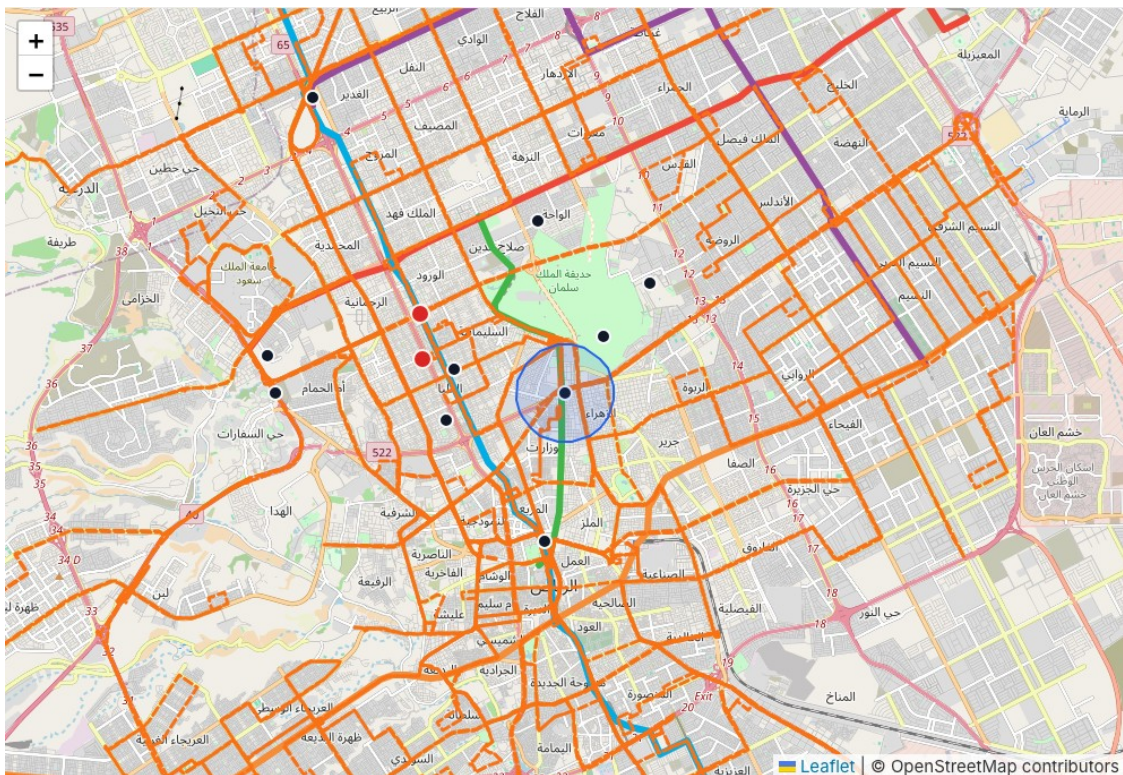
Start the app:

```
python -m uvicorn app.main:app --reload
```

Open the dashboard:

`http://127.0.0.1:8000`

The first page load can take time. Wait for the district selector, layer controls, map container, and score panel area before deciding whether the app is loaded. If Azure Maps is not configured, the app should use the OpenStreetMap fallback and still show the route overlays.



Map layers screenshot

Figure: metro and bus layers rendered locally with OpenStreetMap fallback tiles.

## Local Verification Checklist

- Dashboard loads at `http://127.0.0.1:8000`.
- Map renders using Azure Maps or the OpenStreetMap fallback.
- Metro and bus layers appear.
- District selector has 10 Riyadh districts.
- Selecting a district updates the score panel.
- `/health` returns `{"status":"ok"}`.
- `/api/data-status` reports active sample mode.

Useful checks:

```
curl -s http://127.0.0.1:8000/health
curl -s http://127.0.0.1:8000/api/data-status | python -m json.tool
python -m pytest
python scripts/validate_data.py
```

## 6. Backend Walkthrough

The backend is a small FastAPI application. It serves both the API and the static frontend so the starter kit stays easy to run and easy to deploy.

| Endpoint                                        | Purpose                                         | What a builder should look for                 |
|-------------------------------------------------|-------------------------------------------------|------------------------------------------------|
| GET <code>/health</code>                        | Confirms the app is alive                       | <code>{"status":"ok"}</code>                   |
| GET <code>/api/config</code>                    | Sends runtime settings to the frontend          | app name, map mode, access buffer, data mode   |
| GET <code>/api/routes</code>                    | Returns route summaries and KPI counts          | metro and bus route counts                     |
| GET <code>/api/routes/geojson?mode=metro</code> | Returns map-ready metro geometry                | GeoJSON payload and source                     |
| GET <code>/api/routes/geojson?mode=bus</code>   | Returns map-ready bus geometry                  | GeoJSON payload and source                     |
| GET <code>/api/districts</code>                 | Returns district selector records               | 10 district items in sample mode               |
| GET <code>/api/score?districtId=...</code>      | Returns score and score components              | score, rating, metro count, bus count, penalty |
| GET <code>/api/live-events</code>               | Returns mock or configured event overlays       | delay/event items and source                   |
| GET <code>/api/data-status</code>               | Explains active data source and fallback status | current mode, counts, fallback message         |



## GET /api/data-status

Actual local response captured from http://127.0.0.1:8000

```
{
  "requestedMode": "sample",
  "routes": {
    "count": 106,
    "source": "sample"
  },
  "districts": {
    "source": "sample"
  },
  "liveEvents": {
    "enabled": true,
    "source": "sample",
    "count": 2
  },
  "activeMode": "sample",
  "fallbackMessage": "Azure services are optional. The app falls back to bundled sample files when Blob Storage or Cosmos DB is unavailable."
}
```

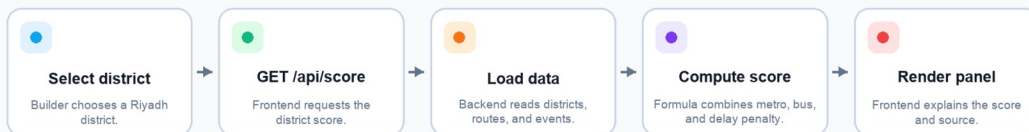
API data status screenshot

Figure: /api/data-status confirms sample mode and explains the fallback path.

## Score Request Flow

### Score request flow

Build path diagram for the Riyadh Mobility Intelligence starter kit



Score request flow

The score formula stays on the backend so it can be tested and reused:

$$\text{score} = (\text{nearby metro count} \times 3) + \text{nearby bus count} - \text{live delay penalty}$$

Keep this formula easy to explain. Teams can later replace the weights or add new inputs such as parking, walkability, public services, heat, or event density.

## 7. Frontend Walkthrough

The frontend is a single-page application built with Vanilla JavaScript. This keeps the starter kit accessible to mixed-experience teams while still showing a complete app workflow.

On startup, `app/static/src/main.js` fetches the core runtime data in parallel:

```
const [config, routes, metro, bus, districts, events, dataStatus] = await Promise.all([
  api.getConfig(),
  api.getRoutes(),
  api.getRouteGeojson("metro"),
  api.getRouteGeojson("bus"),
  api.getDistricts(),
  api.getEvents(),
  api.getDataStatus()
]);
```

```

    api.getLiveEvents(),
    api.getDataStatus(),
  ]);

```

The map layer system makes the city data visible and explainable.

| Layer                | What it shows                        | Why it matters                              |
|----------------------|--------------------------------------|---------------------------------------------|
| Metro lines          | high-capacity mobility corridors     | shows structural transit coverage           |
| Bus routes           | surface transit coverage             | shows fine-grained network reach            |
| Districts            | selectable district points           | anchors the score conversation              |
| Live events          | delay or incident markers            | demonstrates cloud-live extension potential |
| Accessibility buffer | rough selected-district service area | makes the score formula visible             |

District

Central Riyadh



*District selector screenshot*

The score panel translates raw counts into a judge-friendly story:

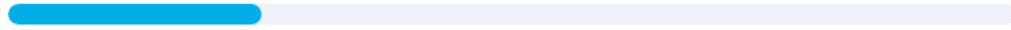
- selected district name
- score number
- rating
- nearby metro count
- nearby bus count
- live-event penalty
- readable formula
- active data source context

## TRANSIT SCORE

Total score: **14** **HIGH** of ~24 max



Nearby metro lines (×3 pts) **1**



Nearby bus routes (×1 pt) **11**



$$\text{score} = (1 \times 3) + 11 - 0 = 14$$

Hackathon-friendly proxy — not a formal transport model.  
Designed to show how open data, cloud storage, and  
simple scoring can support district-level planning  
conversations.

*District score panel screenshot*

*Figure: district score panel after selecting a Riyadh district.*

# Cloud Deployment Blueprint

A credible cloud-live path turns the local starter kit into an Azure-hosted app with map services, data storage, monitoring, and a repeatable teardown plan.

1

**Authenticate**`az login + azd auth login`

2

**Initialize**`azd init and choose subscription/location`

3

**Deploy**`bash scripts/deploy_azure.sh`

4

**Verify**`WEB_APP_URL, /health, /api/data-status`

5

**Operate**`App Insights, logs, fallback status`

6

**Teardown**`destroy_resource_group.sh --yes`

## Resources Created



Container Apps



Container Registry



Azure Maps



Blob Storage



Cosmos DB



App Insights



Log Analytics

## After-Deploy Checks

- Copy WEB\_APP\_URL from azd output
- Open dashboard and wait through slow first load
- Confirm /health returns ok
- Confirm /api/data-status reports active mode
- Use App Insights/logs if startup fails

### Slow-load expectation

Do not judge success too early. Wait for the district selector, layer controls, map area, and score panel before treating the page as failed.

### Cloud-live story for judges

The deployed version uses the same product shape as local mode, then adds Azure Maps, Blob Storage, Cosmos DB, Container Apps, and telemetry so teams can prove a realistic path from starter kit to operational prototype.

# 8. Cloud-Live Deployment

The cloud-live path is the deployable version of the same starter kit. It is useful when a team needs to show that the app can move beyond a laptop and into a credible Azure environment.

## Prerequisites

Before deploying, confirm:

- Azure account is available.
- Azure CLI is installed.
- Azure Developer CLI is installed.
- Shell is authenticated to the right Azure tenant and subscription.
- Target location and resource group are known.
- Local app and tests already pass.

Sign in:

```
az login
azd auth login
azd init
```

Deploy with the helper:

```
bash scripts/deploy_azure.sh
```

Optional location and resource group:

```
bash scripts/deploy_azure.sh eastus rg-riyadh-ud-eastus
```

The helper prints active environment values, then runs the Azure Developer CLI deployment workflow.

## What Deployment Creates

| Azure resource           | Role                                           |
|--------------------------|------------------------------------------------|
| Azure Container Apps     | hosts FastAPI and the static frontend          |
| Azure Container Registry | stores the built container image               |
| Azure Maps account       | supports cloud map services                    |
| Azure Blob Storage       | stores raw and processed mobility files        |
| Azure Cosmos DB          | stores route, district, and event records      |
| Application Insights     | tracks health, latency, failures, and requests |
| Log Analytics            | stores logs and diagnostics                    |

After deployment:

1. Copy `WEB_APP_URL` from azd output.
2. Open the deployed app in a browser.



3. Check <WEB\_APP\_URL>/health.
4. Check <WEB\_APP\_URL>/api/data-status.
5. Confirm the active data mode is expected.
6. Upload processed files to Blob if using Blob mode.
7. Seed Cosmos DB if using Cosmos mode.
8. Inspect App Insights if the deployed app fails or loads slowly.

Optional cloud data steps:

```
python3 scripts/upload_to_blob.py  
PYTHONPATH=. python3 scripts/seed_cosmos.py
```

Teardown when finished:

```
bash scripts/destroy_resource_group.sh --yes
```

The teardown script deletes the current Azure resource group and intentionally requires `--yes`.

# Track Adaptation Routes

Keep the proven shell. Replace the domain data, score logic, explanations, and Azure services to make the track-specific intelligence obvious.

## Transformational Technology

### Mobility Command View

#### Reuse from starter kit

Keep: route layers, event overlay, district selector, KPIs.

#### Make it track-specific

Add: live status, parking demand, peak-load simulation.

## Prosperous People

### District Intelligence

#### Reuse from starter kit

Keep: district selector, score panel, Cosmos-ready records.

#### Make it track-specific

Add: clinics, schools, parks, service-access scoring.

## Sustainable Solutions

### Sustainable Mobility Overlay

#### Reuse from starter kit

Keep: map shell, buffer logic, fallback data path.

#### Make it track-specific

Add: AQI, heat, emissions, clean corridor scoring.

## Culture

### Visitor Movement

#### Reuse from starter kit

Keep: map interaction, route overlays, selected-area storytelling.

#### Make it track-specific

Add: heritage markers, event-day routing, multilingual notes.

Winning pattern: do not rebuild the app. Preserve the local-first API, map shell, fallback chain, and deployment path; make the chosen track unmistakable through data and scoring.

## 9. Starter Kit Adaptation Routes

Treat this codebase as a scaffold. Teams should keep the working shell, replace the domain data, and adapt the score or map layers toward the track they are pursuing.

### Route 1: Mobility Command View

Best fit: Transformational Technology.

Keep:

- route layers
- event overlay
- district selector
- route KPIs
- data-status debugging

Replace or add:

- real congestion or delay source
- parking demand layer
- peak-load simulation
- route delay explanations

Likely Azure services: Azure Maps, Container Apps, Event Hubs, Stream Analytics, Cosmos DB, Application Insights.

### Route 2: District Intelligence

Best fit: Prosperous People.

Keep:

- district selector
- scoring panel
- map focus behavior
- Cosmos-ready district records

Replace or add:

- service access layers such as clinics, schools, parks, or transit stops
- 15-minute city score
- walkability indicators
- district comparison panel

Likely Azure services: Azure Maps, Blob Storage, Cosmos DB, Container Apps, optional Azure OpenAI for plain-language score explanations.

### Route 3: Sustainable Mobility Overlay

Best fit: Sustainable Solutions.

Keep:

- mobility layers
- selected district context
- route scoring pattern
- fallback-aware sample files

Replace or add:

- AQI stations
- heat layer
- pollution hotspot markers
- clean corridor recommendations
- emissions or low-carbon route scoring

Likely Azure services: Azure Maps, Blob Storage, Cosmos DB, Power BI, Application Insights.

## Route 4: Culture and Visitor Movement

Best fit: Culture.

Keep:

- map layer structure
- event markers
- route overlays
- selected place or district panel

Replace or add:

- heritage site markers
- crowd-sensitive route suggestions
- multilingual visitor notes
- event-day access planning

Likely Azure services: Azure Maps, Cosmos DB, Blob Storage, Container Apps, optional translation services.

# 10. Demo Flow

Use this short flow when presenting to mentors or judges.

|  | Step | Show                   | Explain                                                                   |
|--|------|------------------------|---------------------------------------------------------------------------|
|  | 1    | Dashboard landing view | This is a Riyadh mobility intelligence starter kit, not just a static map |
|  | 2    | Metro and bus toggles  | The map layers show public transport coverage                             |
|  | 3    | District selector      | District context drives the scoring workflow                              |
|  | 4    | Score panel            | The score is transparent and easy to adapt                                |

| Step | Show                   | Explain                                                       |
|------|------------------------|---------------------------------------------------------------|
| 5    | /api/data-status       | The app is fallback-aware and cloud-live ready                |
| 6    | Azure workflow diagram | Services map to hosting, maps, files, records, and monitoring |
| 7    | Adaptation routes      | The same scaffold supports multiple hackathon tracks          |

Keep the live demo local unless the deployed app has already passed /health and /api/data-status checks.

# Appendix

## Common Commands

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
cp .env.example .env
python -m uvicorn app.main:app --reload
python -m pytest
python scripts/validate_data.py
bash scripts/deploy_azure.sh
bash scripts/destroy_resource_group.sh --yes
```

## Debugging Checklist

- If the page loads slowly, wait for the district selector, layer controls, map area, and score panel before judging success.
- If the map is blank, clear AZURE\_MAPS\_KEY in .env and reload so the local fallback can be used.
- If cloud data is missing, open /api/data-status first.
- If sample files fail, run python scripts/validate\_data.py.
- If the deployed app fails, check /health, /api/data-status, Container App logs, and Application Insights.

## Visual Asset Checklist

| Asset                        | Included in this guide                            |
|------------------------------|---------------------------------------------------|
| app layer icon row           | docs/assets/rebuild-guide/icon-row-app-layers.png |
| four-track icon row          | docs/assets/rebuild-guide/icon-row-tracks.png     |
| local dashboard screenshot   | docs/assets/rebuild-guide/local-dashboard.png     |
| map layer screenshot         | docs/assets/rebuild-guide/map-layers.png          |
| district selector screenshot | docs/assets/rebuild-guide/district-               |

| Asset                           | Included in this guide                                           |
|---------------------------------|------------------------------------------------------------------|
|                                 | <code>selector.png</code>                                        |
| district score panel screenshot | <code>docs/assets/rebuild-guide/district-score-panel.png</code>  |
| data-status endpoint screenshot | <code>docs/assets/rebuild-guide/api-data-status.png</code>       |
| local workflow diagram          | <code>docs/assets/rebuild-guide/diagram-local-first.png</code>   |
| Azure workflow diagram          | <code>docs/assets/rebuild-guide/diagram-azure-backed.png</code>  |
| fallback chain diagram          | <code>docs/assets/rebuild-guide/diagram-data-fallback.png</code> |
| score request diagram           | <code>docs/assets/rebuild-guide/diagram-score-request.png</code> |

## Resource Index

Use this page as the resource hub instead of repeating links in the footer. Keep the cover page focused on the repo, local app, and deployed app URL; keep this appendix focused on build, deployment, and Azure references.

| Need                         | Open                                                                                                                                  |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| GitHub repo                  | <a href="https://github.com/Esturban/microsoft-hackathon-riyadh-mobility">github.com/Esturban/microsoft-hackathon-riyadh-mobility</a> |
| Local dashboard              | <a href="http://127.0.0.1:8000">http://127.0.0.1:8000</a>                                                                             |
| atomcamp Arabia              | <a href="https://atomcamparabia.com">atomcamparabia.com</a>                                                                           |
| Azure Developer CLI docs     | <a href="#">Microsoft Learn</a>                                                                                                       |
| <code>azd up</code> workflow | <a href="#">Microsoft Learn</a>                                                                                                       |
| Azure Container Apps docs    | <a href="#">Microsoft Learn</a>                                                                                                       |
| Azure Maps docs              | <a href="#">Microsoft Learn</a>                                                                                                       |
| Azure Cosmos DB docs         | <a href="#">Microsoft Learn</a>                                                                                                       |
| Student quickstart           | <code>README.md</code>                                                                                                                |
| Architecture notes           | <code>docs/architecture.md</code>                                                                                                     |



| Need                   | Open                                        |
|------------------------|---------------------------------------------|
| Azure deployment notes | <a href="#">docs/azure_deployment.md</a>    |
| Data source notes      | <a href="#">docs/data_sources.md</a>        |
| Troubleshooting        | <a href="#">docs/troubleshooting.md</a>     |
| Demo script            | <a href="#">docs/judging_demo_script.md</a> |

## Glossary

### Azure-backed live path

The deployable path where the starter kit uses Azure services for maps, hosting, files, records, and monitoring.

### Blob Storage

Azure file storage for raw and processed mobility datasets.

### Cosmos DB

Azure document database for route summaries, district records, and event records.

### FastAPI

Python web framework used to serve the API and static frontend.

### GeoJSON

JSON format for map features such as points, lines, and polygons.

### Mobility Intelligence Starter Kit

The reusable scaffold that teams can adapt into mobility, district, sustainability, or culture prototypes.

### Sample fallback

The local files that keep the app useful even when cloud services or remote data are unavailable.

### Single-page application

The current frontend shape: one main web page that loads data and updates panels dynamically.

## Closing Note

The starter kit is strongest when it stays practical: one useful local screen, clear route layers, transparent scoring, and a cloud-live path that is credible without becoming heavy. Build the smallest version that can be explained well, then extend it toward the hackathon track that matters most.